DAY - 2 SEAT NUMBER M 0 3 5 4 8 4

2011	II	24	1500	V – 334	(E)

COMPUTER SCIENCE
PAPER - I (D-9)

Time : 3 Hours	4 Pages	Max. Marks : 50

Instructions : *(1) All questions are compulsory.*

(2) Figures to the right indicate full marks.

(3) Use of calculator is not allowed.

(4) Draw neat diagrams wherever necessary.

(5) Due credit will be given for program with appropriate remarks or comments.

1. (A) Select correct alternatives and rewrite the following statements :

(a) A program under execution, which competes for the CPU time and other resources is called _____. 1

(i) Process

(ii) Multi-programming

(iii) Context Switching

(iv) None of these

(b) Accessing each element only once is called _____. 1

(i) Traversing

(ii) Searching

(iii) Inserting

(iv) Deleting

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(c) Ability to take more than one form is called _____ 1

(i) Inheritance

(ii) Polymorphism

(iii) Classes

(iv) None of these

(d) To place image into an HTML File _____ attribute is used in IMG tag. 1

(i) <URL>

(ii) <ALT>

(iii) <SRC>

(iv) <HREF>

(B) Answer **any two** of the following :

(a) Explain in short the function of Menu Bar and Scroll Bar components of GUI. 3

(b) In case of Data Structure, explain the following terms : 3

(i) Linear Array

(ii) Linked List

(iii) Tree

(c) Explain the following HTML tags with suitable examples : 3

(i) <P>

(ii)

(iii) <HR>

2. (A) Answer **any two** of the following :

(a) Differentiate between Traditional Procedural Programming Approach and Object Oriented Programming Approach. 3

(b) What is Binary Tree ? Explain terminology describing family relationship between the elements of a binary tree with suitable example. 3

(c) Write a Bubble Sort Algorithm with suitable example. 3

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(B) Answer **any one** of the following :

(a) Explain the following terms : 4

(i) Page Fault

(ii) Working Set

(iii) Page Replacement Policy

(iv) Dirty Page

(b) What is Class ? Explain general form of Class Declaration. 4

3. (A) Answer **any two** of the following :

(a) What is Pointer ? What are the advantages of Pointer in C++ ? 3

(b) What is Computer Virus ? State any four methods by which Virus can infect other programs. 3

(c) What is Constructor and Destructor ? State difference between them. 3

(B) Answer **any one** of the following :

(a) With reference to Process Management, explain the terms : 4

(i) External Priority

(ii) Purchased Priority

(iii) Internal Priority

(iv) Time Slice

(b) What is Record ? Show representation of records in memory by considering suitable example of three records and three fields. 4

4. (A) Answer **any two** of the following :

(a) Write an algorithm for inserting an element to a Linear Array. 3

(b) Explain with the help of memory map diagram of Single User Operating System. 3

(c) Explain in brief the three special characteristics of Static Data Member in a Class. 3

(B) Answer **any one** of the following :

(a) What is Operator Function ? Describe syntax of an Operator Function. Explain difference between Operation Function as a Member Function and as a Friend Function. 4

(b) Explain Paging and Segmentation. 4

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5. Answer **any two** of the following :

(a) Write a C++ program to accept a number and test whether it is prime or not. 5

(b) Write an Object Oriented Program in C++ to read a set of 10 numbers and store it as an one dimensional array; again read a number 'd' and check whether the number 'd' is present in the array, if it is so, print out how many times the number 'd' is repeated in the array. 5

(c) Write a HTML Code for the following output : 5

COMPUTER DEVICES ← Text size is h_2, align Center

- INPUT DEVICES
  - (1) Key Board
  - (2) Mouse
- STORAGE DEVICES
  - (1) Hard Disk
  - (2) Floppy Disk
  - (3) Compact Dist
- OUTPUT DEVICES
  - (1) Screen
  - (2) Printer

OR

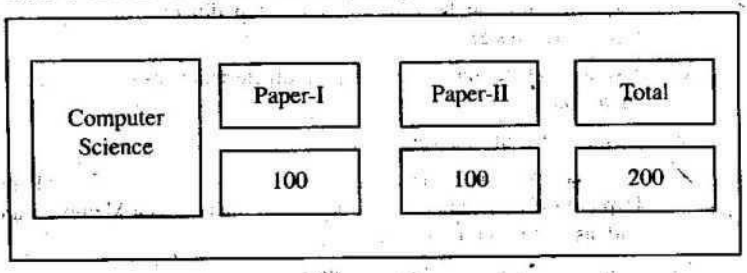
5. Answer **any two** of the following :

(a) Write a program in C++ to find Greatest Common Divisor (GCD) of two natural numbers. 5

(b) Write an Object Oriented Program in C++ to read an integer number and find out the sum of all the digits of a number. 5

(For eg. n = 1256, SUM = 1 + 2 + 5 + 6 = 14)

(c) Write a HTML Code for the following : 5



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